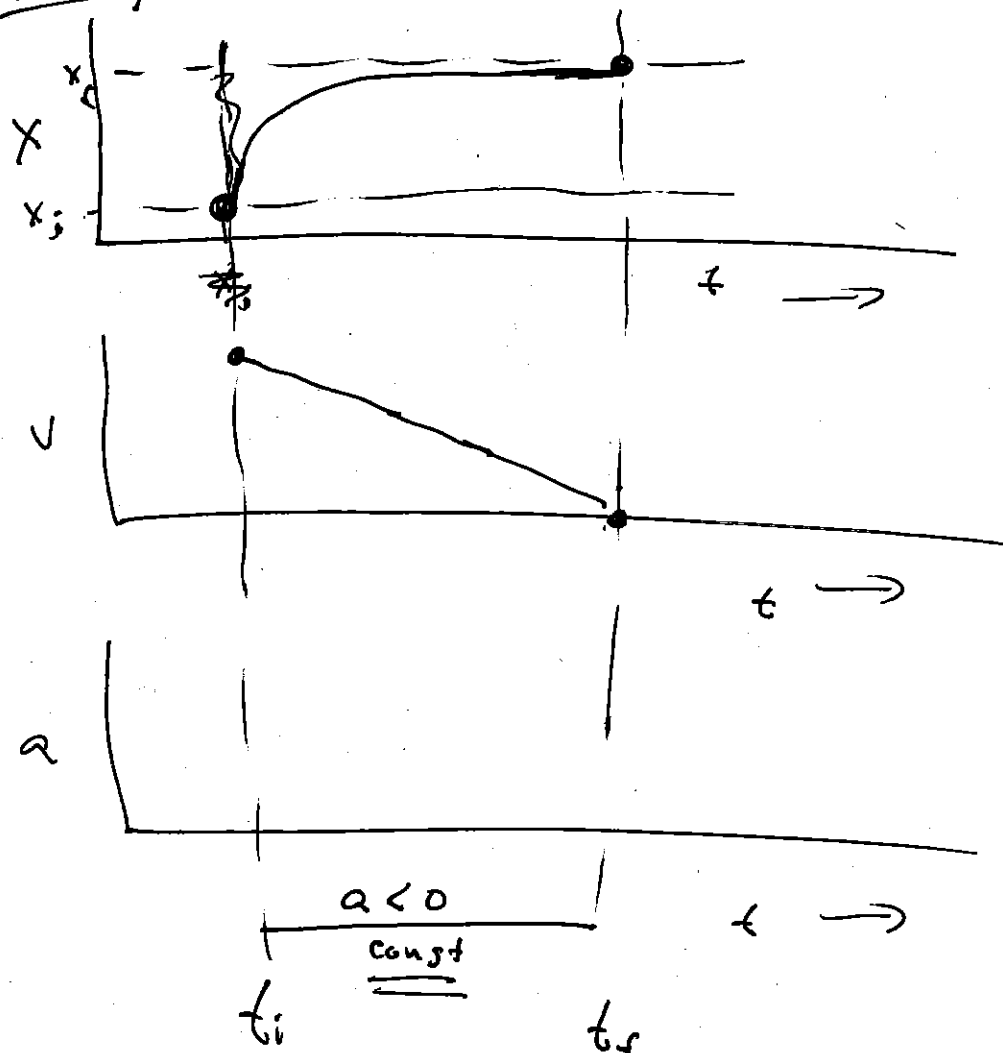


$x, v, a/t$


plane lands
on aircraft

carrier
 $v_i = 70 \text{ m/s}$

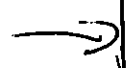
$v_f = 0$

$\Delta x = 100 \text{ m}$

use position velocity accel graphs
to understand problems

$$v_{ave} = \frac{x_f - x_i}{t} \quad \text{or} \quad v_{ave} = \frac{v_i + v_f}{2}$$

$$\frac{x_f - x_i}{t} = \frac{v_i + v_f}{2}$$



$$x_f = x_i + \frac{v_i + v_f}{2} t$$

EQ 1

a_{ave} (all a 's are average)

2/9

restricting ourselves mathematically

$$a = \frac{v_f - v_i}{t} \rightarrow \boxed{v_f = v_i + at} \quad \text{EQ 2}$$

Brd Eq

$$1 \quad x_f = x_i + \frac{v_i + v_f}{2} t \quad \leftarrow$$

$$2 \quad v_f = v_i + at \rightarrow \boxed{t = \frac{v_f - v_i}{a}}$$

$$x_f = x_i + \frac{v_i + v_f}{2} \cdot \frac{v_f - v_i}{a}$$

$$x_f = x_i + \frac{v_i v_f - v_i^2 + v_f^2 - v_i v_f}{2a}$$

$$x_f = x_i + \frac{v_f^2 - v_i^2}{2a}$$

$$v_f^2 - v_i^2 = 2a(x_f - x_i)$$

$$\boxed{v_f^2 = v_i^2 + 2a(x_f - x_i)}$$

EQ 3

4th EQ

$$1 \quad X_f = X_i + \frac{v_i + v_f}{2} t$$

$$2 \quad v_f = v_i + at \rightarrow$$

3/9

$$X_f = X_i + \frac{v_i + v_i + at}{2} t$$

$$X_f = X_i + \frac{2v_i + at}{2} t$$

$$X_f = X_i + v_i t + \frac{1}{2} at^2$$

EQ 4

NOW EXAMPLES OF USING THE
4 Kinematic Equations

$$X_f = X_i + \frac{v_i + v_f}{2} t$$

$$v_f = v_i + at$$

$$v_f^2 = v_i^2 + 2a(X_f - X_i)$$

$$X_f = X_i + v_i t + \frac{1}{2} at^2$$

4/9

2 cars 1000 m apart approach each other.
Initially at rest. When do they collide?

car 1 $x_{1i} = 0$ $a_1 = 3 \text{ m/s}^2$

car 2 $x_{2i} = 1000 \text{ m}$ $a_2 = -2 \text{ m/s}^2$

$$x_{1f} = \cancel{x_{1i}} + \cancel{v_{1i} t} + \frac{1}{2} a_1 t^2$$

$$x_{1f} = \frac{1}{2} a_1 t^2$$

$$x_{2f} = x_{2i} + \cancel{v_{2i} t} + \frac{1}{2} a_2 t^2$$

$$x_{2f} = x_{2i} + \frac{1}{2} a_2 t^2$$

"Collide" means $x_1 = x_2$

$$\frac{1}{2} a_1 t^2 = x_{2i} + \frac{1}{2} a_2 t^2$$

$$x_{2i} = \frac{1}{2} (a_1 - a_2) t^2$$

$$t = \sqrt{\frac{2 x_{2i}}{a_1 - a_2}} = \sqrt{\frac{2 \cdot 1000 \text{ m}}{3 - (-2) \text{ m/s}^2}}$$

$$= \sqrt{\frac{2000 \text{ m}}{5 \text{ m/s}^2}} = \sqrt{400 \text{ s}^2} = \boxed{20 \text{ s}}$$

where? $x_{1f} = \frac{1}{2} a_1 t^2 = \frac{1}{2} 3 \frac{\text{m}}{\text{s}^2} (20 \text{ s})^2 = \boxed{600 \text{ m}}$

Plane at 70 m/s lands on 100 m carrier $5/9$
 $a = ?$

$$v_i = 70 \text{ m/s} \quad (v_f = 0)$$

$$\Delta x = 100 \text{ m}$$

$$v_f^2 = v_i^2 + 2a(x_f - x_i)$$

$$a = \frac{v_f^2 - v_i^2}{2\Delta x} = \frac{0 - v_i^2}{2\Delta x}$$

$$= \frac{-70^2}{2 \cdot 100} \frac{(\text{m/s})^2}{\text{m}}$$

$$= \frac{-4900}{200} \text{ m/s}^2$$

$$= -24.5 \text{ m/s}^2 \quad 2\frac{1}{2} \text{ "gees"}$$

Wile E Coyote falls from 100 m cliff

$h = 100 \text{ m}$

$a_y \downarrow = -g$ how long to fall? $v_{y,i} = 0$

$$y_f = y_i + v_{y,i}t + \frac{1}{2}a_y t^2$$

$$0 = h + 0 - \frac{gt^2}{2}$$

$$t = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2 \cdot 100 \text{ m}}{9.8 \text{ m/s}^2}} = 4.52 \text{ s}$$

$y^+ \uparrow$

$$v_{yf} = ?$$

$$v_f^2 = v_i^2 + 2a(x_f - x_i)$$

$$v_{yf}^2 = v_{yi}^2 + 2a_y(y_f - y_i)$$

6/9

$$v_{yf}^2 = -2g(0 - h)$$

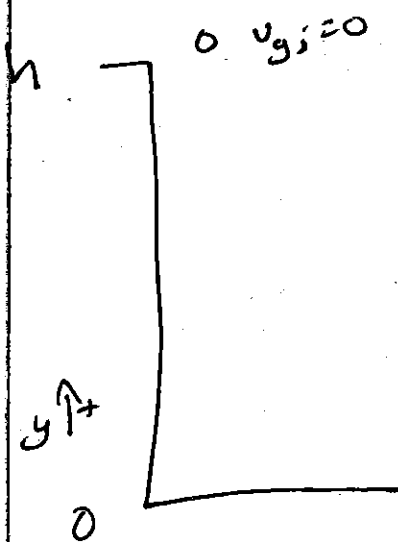
$$v_{yf}^2 = 2gh$$

$$v_{yf} = \sqrt{2gh} = \sqrt{2 \cdot 9.8 \frac{\text{m}}{\text{s}^2} \cdot 100\text{m}}$$

$$v_{yf} = \sqrt{1960 \frac{\text{m}^2}{\text{s}^2}} = -44.27 \frac{\text{m}}{\text{s}}$$

- chosen !

A penny is dropped down a well. It hits the water at 2s. How deep?



$$y_i = h = ?$$

$$v_{yi} = 0$$

$$y_f = 0$$

$$a_y = -g$$

$$t = 2\text{s}$$

$$y_f = y_i + v_{yi}t + \frac{1}{2}a_yt^2$$

$$0 = h + 0 - \frac{g}{2}t^2$$

$$h = \frac{g}{2}t^2$$

$$= \frac{9.8 \frac{\text{m}}{\text{s}^2} (2\text{s})^2}{2} = \boxed{19.6 \text{ m}}$$

7/9
A penny is thrown down a well at

-2 m/s (note thrown down means $v < 0$)

Hits in 1 s. How deep

$$y_i = h$$

$$y_f = 0$$

$$v_{y_i} = -2 \text{ m/s}$$

$$t = 1 \text{ s}$$

$$y_f = y_i + v_{y_i} t + \frac{1}{2} a_y t^2$$

$$0 = h + v_{y_i} t - \frac{g}{2} t^2$$

$$h = \frac{g}{2} t^2 - v_{y_i} t$$

$$h = \frac{9.8 \text{ m/s}^2}{2} (1 \text{ s})^2 - (-2 \text{ m/s})(1 \text{ s})$$

$$h = 4.9 \text{ m} - (-2 \text{ m})$$

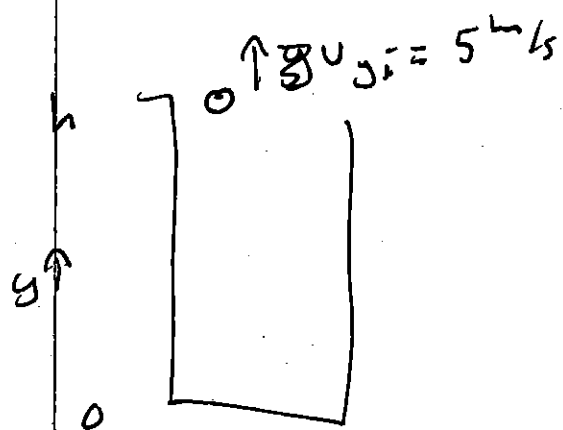
$$= 4.9 + 2 \text{ m}$$

$$\boxed{= 6.9 \text{ m}}$$

Penny is thrown upward at 5 m/s

8/9

It goes up, falls down, and hits water
in ~~3~~ 3 s . How deep?



$$y_f = 0$$

$$y_i = h \text{ ?}$$

$$v_{yi} = +5 \text{ m/s}$$

$$t = 3 \text{ s}$$

$$a_y = -g$$

$$y_f = y_i + v_{yi}t + \frac{1}{2}a_yt^2$$

$$0 = h + v_{yi}t - \frac{g}{2}t^2$$

$$h = \cancel{v_{yi}t} - \frac{g}{2}t^2 + v_{yi}t$$

$$h = \frac{9.8 \text{ m/s}^2}{2} (3 \text{ s})^2 - 5 \text{ m/s} \cdot 3 \text{ s}$$

$$= \frac{9.8}{2} \cdot 9 - 15 \text{ m}$$

$$= 44.1 - 15 \text{ m}$$

$$= 29.1 \text{ m}$$

Penny is dropped into 20 m
deep well. When does it hit?

9/9

$$y_f = 0 \quad y_i = 20 \text{ m} \quad v_{y,i} = 0 \quad a_y = -g$$

$$y_f = y_i + v_{y,i} t + \frac{1}{2} a_y t^2$$

$$0 = h + 0 - \frac{g}{2} t^2$$

$$h = \frac{g}{2} t^2$$

$$t = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2 \cdot 20 \text{ m}}{9.8 \text{ m/s}^2}} = 2.02 \text{ s}$$